

CAT: preserve the intended spaces

Ask for the behavior, not just a neat result

01 / THE SAS EXAMPLE

```
data test;
  length first middle last $8
         name1 name2 $24;
  input first $ middle $ last $;
  name1 = cat(first, middle, last);
  name2 = catx(' ', first, middle, last);
  datalines;
John Q Public
Jane . Doe
Bob . Smith
;
run;

proc print data=test noobs;
run;
```

REPORTED TEST SETUP / 21 SEP 2026
Claude Sonnet 4.5 / AWS Bedrock GovCloud
Claude Code 2.1.278 / effortLevel=high / prompt v0.1.0

ATTACH SOURCE OR PASTE IT AFTER THE PROMPT. START FRESH EACH TIME.

WHAT YOU WILL PRACTICE

Distinguish preserving source behavior from silently cleaning the data.

CAT keeps blanks in these character fields. CATX trims them and separates nonblank values. Both behaviors are intentional. Inspect the spaces as well as the readable names.

EXPECTED RESULT

Row	CAT: name1	CATX: name2
1	John____Q____Public__	John Q Public
2	Jane_____Doe_____	Jane Doe
3	Bob_____Smith____	Bob Smith

Each underscore represents one stored blank in name1. name2 hides trailing storage padding. Input periods become blank middle fields. Both stored result columns have width 24.

REPORTED FULL-REQUIREMENT MATCHES
Prompt A 0/3 | Prompt B 3/3

Reference values checked in Python; SAS execution pending.

Prompt A: leave choices open

High degrees of freedom / Notice the decisions the model makes for you.

COPY THIS PROMPT + THE SAS EXAMPLE

Convert this SAS code to Python.

Show code in this response. Do not create or run files, or read existing Python files.

REPORTED TRIAL RESULTS / 21 SEP 2026

Readable names can hide missing spaces

Run	CATX approach	Full match
A1	Join with spaces	No
A2	Two alternatives shown	No
A3	Filter periods + join	No

All three used unpadded + concatenation for CAT.

A3 matched readable CATX names; stored widths were missing.

A2 offered both CATX recipes seen in A1 and A3.

SOURCES: results-20260921.csv; trial-log-20260921.md; analysis-20260921.md

WHY CHOICES REMAIN OPEN

- 1

CAT keeps the stored blanks
All A runs reportedly joined raw inputs. The + operator is valid Python, but it cannot restore omitted padding.
- 2

A readable name is only one check
A3 filtered the input periods and got the readable names. Both stored results still needed width 24.
- 3

Alternatives need a decision
A2 offered two CATX recipes. Compare each with the required behavior before accepting either.

ANTI-PATTERN: ACCEPTING A NEAT NAME

Inspect visible blanks and stored widths. Ask for cleanup separately after checking the equivalent conversion.

Different Python methods can preserve the same behavior.

Prompt B: make the choices explicit

Low degrees of freedom / State the behavior, then ask for a check.

COPY THIS PROMPT + THE SAME SAS EXAMPLE

```
Convert this SAS CAT/CATX code to Python using pandas.
- Use first, middle, last in the original row order.
  Map the input periods to blank middle fields, not literal dots.
- Right-pad every input field to 8 characters with spaces.
- name1: concatenate the padded fields without trimming (CAT).
- name2: strip ordinary leading/trailing spaces from each field,
  skip blank fields, and join with one space (CATX).
- Store both result columns padded to width 24.
- Include assertions for both widths and all results below.
  For name1 checks, _ denotes one space:
  John__Q____Public__
  Jane____Doe____
  Bob____Smith__
  name2, with trailing padding hidden:
  John Q Public; Jane Doe; Bob Smith.
- Print name1 with spaces visible and name2 in readable form.
  Keep stored values unchanged; explain the difference briefly.

Show code in this response. Do not create or run files,
or read existing Python files.
```

SAS REFERENCE

Reports describe six fresh parallel agents; complete transcripts and execution logs are absent.
Passing assertions are reported. Local reference checked; SAS run pending. Three runs do not guarantee future results.

WHY THESE DETAILS HELP

- 1 Separate CAT from CATX**
CAT preserves padded fields. CATX strips ordinary spaces, skips blanks and joins with one space.
- 2 Define the stored values**
Map input periods to blanks. Require input widths of 8 and result widths of 24; preserve row order.
- 3 Request a checkable result**
Name exact values and widths to assert. B1-B3 report padded CAT and strip/filter/join CATX.

REPORTED RESULTS / THREE B RUNS

Run	Full match	Assertions
B1	Yes	Passed
B2	Yes	Passed
B3	Yes	Passed

CHECK IT YOURSELF
Check exact CAT spaces and widths of 8 / 24.
Blank middle fields must add no CATX separator.

One behavior, three implementations

One behavior contract, three implementations. Run the companion files and the same three acceptance tests against shared fixture data.

SAS / SOURCE BEHAVIOR

```
%macro translate(in=, out=);
data &out;
  length first middle last $8
         name1 name2 $24;
  set &in;
  name1 = cat(first, middle, last);
  name2 = catx(' ', first, middle, last);
run;
%mend;
```

Input fields have width 8 and both outputs width 24. CAT preserves blanks; CATX trims outer blanks and skips blank fields.

PYTHON / PANDAS

```
def translate(df):
    out = df.copy(deep=True)
    cols = ["first", "middle", "last"]
    for col in cols:
        out[col] = out[col].str.ljust(8)
    out["name1"] = (
        out["first"] + out["middle"]
        + out["last"])
    out["name2"] = out[cols].apply(
        catx_space, axis=1).str.ljust(24)
    return out
```

catx_space helper in reference.py strips ordinary spaces, drops blank parts and joins with one space. It does not trim CAT.

R / BASE

```
translate <- function(df) {
  out <- df
  cols <- c("first", "middle", "last")
  for (col in cols)
    out[[col]] <- sprintf(
      "%-8s", out[[col]])
  out$name1 <- paste0(
    out$first, out$middle, out$last)
  a <- apply(out[cols], 1, catx_space)
  out$name2 <- sprintf("%-24s", a)
  out
}
```

catx_space helper in reference.R uses trimws(..., whitespace=" "), removes empty parts, then paste(..., collapse=" ").

CAT and CATX have different jobs
CAT joins the three padded fields exactly. CATX trims ordinary outer spaces, omits blank fields and inserts one separator between retained fields.

Missing belongs to input decoding
CSV blank fields represent the original missing character values. Original unquoted input periods are decoded before translation; stored text is not blindly cleaned.

SAS CAT family

pandas ljust

R trimws

Three checks that earn the pass

Use the same inputs and expected behavior in each language. Check the contract as well as the values.

T1 / Original three-row fixture

INPUT / CONDITION

John/Q/Public, Jane/blank/Doe and Bob/blank/Smith.

EXPECTED

Exact CAT strings preserve 8+8+8 positions; padded CATX values read John Q Public, Jane Doe and Bob Smith.

WHAT THIS CATCHES

Dropping CAT padding, inserting separators for blank middle fields, or changing the readable CATX names.

T2 / Added blank/space boundary

INPUT / CONDITION

Added outer spaces, all blank fields, internal space Mary Ann, and literal tabs.

EXPECTED

CAT preserves all stored positions; CATX omits empty fields and retains internal spaces and tabs.

WHAT THIS CATCHES

Broad whitespace stripping, doubled separators, or treating an all-blank CATX result as missing numeric data.

T3 / Preservation, widths and order

INPUT / CONDITION

All shared rows reversed; original caller data saved.

EXPECTED

Caller unchanged; input widths 8 and output widths 24; exact stored values and row order retained.

WHAT THIS CATCHES

Readable-name-only checks, missing output padding, input mutation or row reordering.

REFERENCE CODE + TESTS: `module-04-cat/appendix/`

Run: `python tools/run_appendix_tests.py --modules 04`

New reference code, separate from historical trials. SAS checks await execution; scope is stated in the companion README.